UPPSALA UNIVERSITET

STUDENT

1D039-0010-BEX

TENTAMEN

1DL301 Databasteknik I

Kurskod	1DL301
Bedömningsform	--
Starttid	12.04.2025 06:00
Sluttid	12.04.2025 11:00
Bedömningsfrist	--
PDF skapad	03.01.2026 14:12

Instructions

Uppgift	Uppgiftstitel	Status	Poäng	Uppgiftstyp
☑	Instructions	Obesvarad		Formulär
1	♣ Question G1: When	Rätt	0/0	Flervalsfråga
2	♣ Question G2: How many	Rätt	0/0	Flervalsfråga
3	♣ Question G3: Study program	Fel	0/0	Flervalsfråga

Misc ER

Uppgift	Uppgiftstitel	Status	Poäng	Uppgiftstyp
4	♥ Interpreting ER model	Rätt	1/1	Flersvarsfråga

ER to relational

Uppgift	Uppgiftstitel	Status	Poäng	Uppgiftstyp
5	Mapping ER to relational model	Rätt	1/1	Flervalsfråga

EER

Uppgift	Uppgiftstitel	Status	Poäng	Uppgiftstyp
6	EER to Relational	Fel	0/1	Flervalsfråga
7	ER to relational model	Rätt	1/1	Flervalsfråga

SQL

Uppgift	Uppgiftstitel	Status	Poäng	Uppgiftstyp
13	SQL	Rätt	1/1	Flervalsfråga
9	SQL	Rätt	1/1	Flervalsfråga

8	SQL	Rätt	1/1	Flervalsfråga
19	SQL	Rätt	1/1	Flervalsfråga
12	SQL	Rätt	1/1	Flervalsfråga

SQL 2

Uppgift	Uppgiftstitel	Status	Poäng	Uppgiftstyp
24	SQL B1	Rätt	1/1	Flervalsfråga
20	SQL B1	Rätt	1/1	Flervalsfråga
21	SQL B1	Rätt	1/1	Flervalsfråga
22	SQL B1	Rätt	1/1	Flervalsfråga

Section 8

Uppgift	Uppgiftstitel	Status	Poäng	Uppgiftstyp
26	♥ Functional Dependencies	Rätt	1/1	Flersvarsfråga

Normalization-Functional dependencies

Uppgift	Uppgiftstitel	Status	Poäng	Uppgiftstyp
28	Normal Forms Q2 (DBI 2022)	Rätt	1/1	Flervalsfråga
27	Normal Forms Q1	Rätt	1/1	Flervalsfråga

Transactions Schedule

Uppgift	Uppgiftstitel	Status	Poäng	Uppgiftstyp
30	♥ Serializability	Rätt	1/1	Flersvarsfråga

Transactions - Precedence graphs

Uppgift	Uppgiftstitel	Status	Poäng	Uppgiftstyp
32	Transactions- Precedence graphs	Rätt	1/1	Flervalsfråga

Transactions

Uppgift	Uppgiftstitel	Status	Poäng	Uppgiftstyp
33	Recoverability	Rätt	1/1	Flervalsfråga

Relational Model Constraints

Uppgift	Uppgiftstitel	Status	Poäng	Uppgiftstyp
34	♥ Constraint violations	Rätt	1/1	Flersvarsfråga
35	♥ Constraint violations	Rätt	1/1	Flersvarsfråga

Section 12

Uppgift	Uppgiftstitel	Status	Poäng	Uppgiftstyp
39	Multilevel Security	Rätt	1/1	Flervalsfråga

Miscellaneous

Uppgift	Uppgiftstitel	Status	Poäng	Uppgiftstyp
40	SQL Injection	Rätt	1/1	Flervalsfråga
41	Choosing the right index	Rätt	1/1	Flervalsfråga

Instructions

Uppsala University

Department of Information Technology

Database Design I (1DL301)

April 2025

Instructions: Read through the complete exam and note below any unclear directives before you start solving the questions. Answer **all** questions.
The paper has two types of questions:

- If a **question is marked with ♥** you must **select ALL correct choices**. If you do not select all correct choices or you include any incorrect choice, your answer will be marked as incorrect.
- For all **other questions** you must **select only one choice** even if there are several correct choices. Your answer will be marked as correct if you select any of the correct choices. If you select an incorrect choice or select more than one choice, your answer will be marked as incorrect.

Please also answer questions: ♣ Q1, Q2 and Q3 which can be useful to us.

Grading. For each correct answer, you gain 1 point. A wrong answer does not generate negative points. To achieve a grade of 3, you must gain at least 14 points in the whole exam. To achieve a grade of 4, you must gain at least 17 points in the whole exam. To achieve a grade of 5, you must collect at least 21 points in the whole exam.

If you find any unclear directives, please note the question number below and explain what you think is unclear.

1 ♣ **Question G1: When**

General questions (useful for us)

When have you attended the course?

Select one alternative (no points awarded for this question):

- (●) 2024 ✔
- ( ) 2023
- ( ) 2022
- ( ) Before 2022

2 ♣ **Question G2: How many**

General questions (useful for us)

How many lectures have you attended?

Select one alternative (no points awarded for this question):

- (●) None or very few ✔
- ( ) Around 25%
- ( ) Around 50%
- ( ) Around 75%
- ( ) Almost all

3 ♣ **Question G3: Study program**

General questions (useful for us)

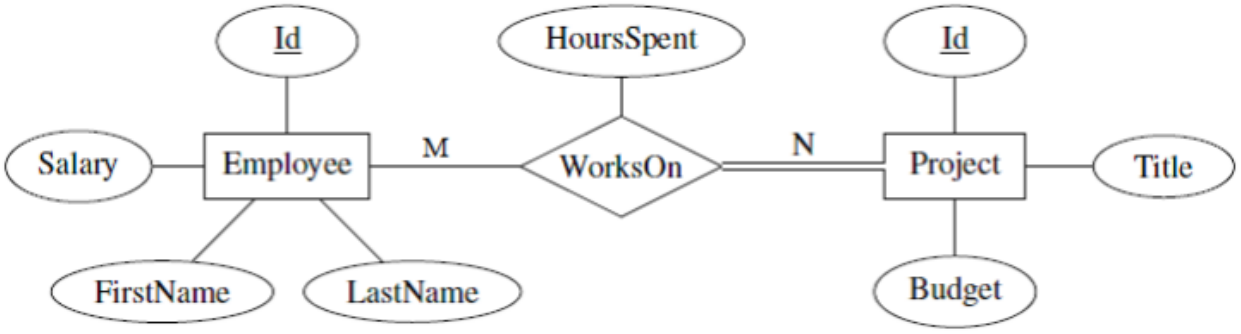
What is your study program?

Select one alternative (no points awarded for this question):

- F ✔
- STS
- CS
- X
- IT ✖
- MIA
- Data Science
- None of the previous answers

4 ♥ Interpreting ER model

For the depicted ER model, select **all** statements that must hold!

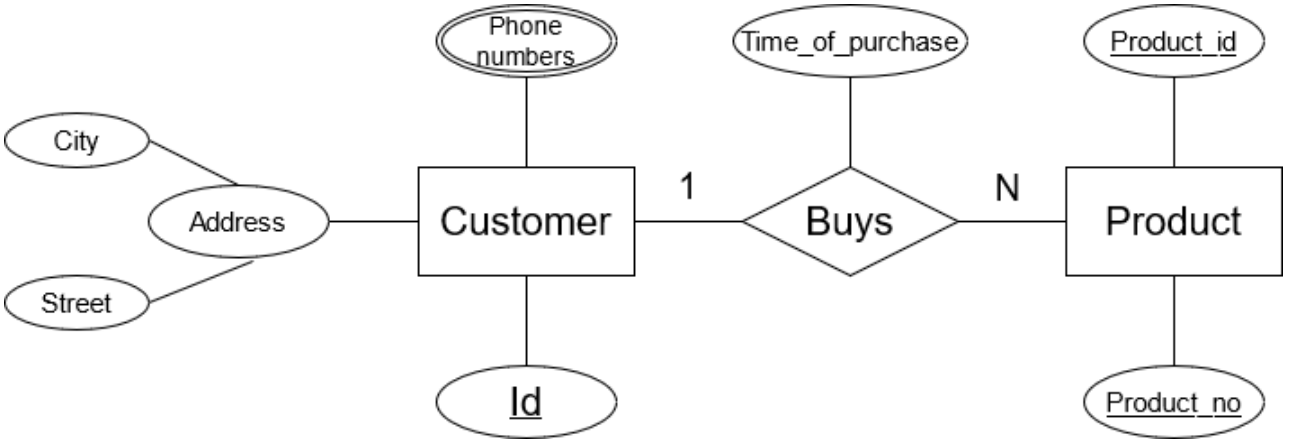


♥ Select one or more alternatives:

- [x] An employee can work on several projects.
- [x] There can be employees who work on no projects.
- [x] Many employees can work on a specific project.
- [ ] Every employee has a unique combination of first and last name.
- [ ] There can be projects on which no employees work.

5 Mapping ER to relational model

Which of the following options is a correct mapping of the ER diagram to the relational model?



Select one alternative:

- CUSTOMER(Id, {Phone numbers}, Address(City, Street)), PRODUCT(Product_id, Product_no, Customer_id, Time_of_purchase), with Cutomer_id $^{FK}\rightarrow$ CUSTOMER(Id)

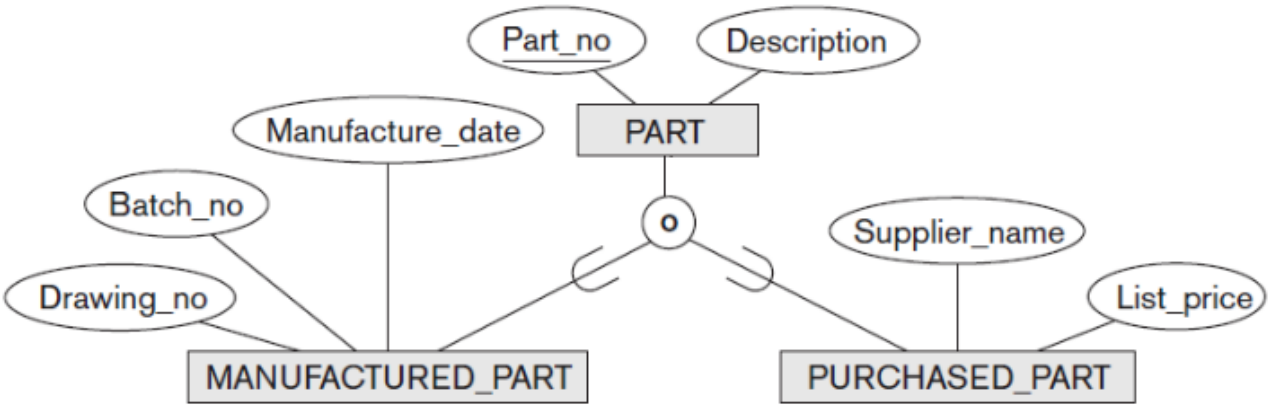
- CUSTOMER(Id, City, Street, Product_id, Product_no), with {Product_id, Product_no} $^{FK}\rightarrow$ PRODUCT({Product_id, Product_no}) PHONES(Phone_number, Id), with Id $^{FK}\rightarrow$ CUSTOMER(Id), PRODUCT(Product_id, Product_no, Time_of_purchase)

- CUSTOMER(Id, {Phone numbers}, Address, City, Street), BUYS(Id, Product_id, Time_of_purchase), with Id $^{FK}\rightarrow$ CUSTOMER(Id) and Product_id $^{FK}\rightarrow$ PRODUCT(Product_id), PRODUCT(Product_id, Product_no)

- CUSTOMER(Id, City, Street), PHONES(Phone_number, Id), with Id $^{FK}\rightarrow$ CUSTOMER(Id), PRODUCT(Product_id, Product_no, Time_of_purchase, Buyer_id), with Buyer_id $^{FK}\rightarrow$ CUSTOMER(Id)

6 EER to Relational

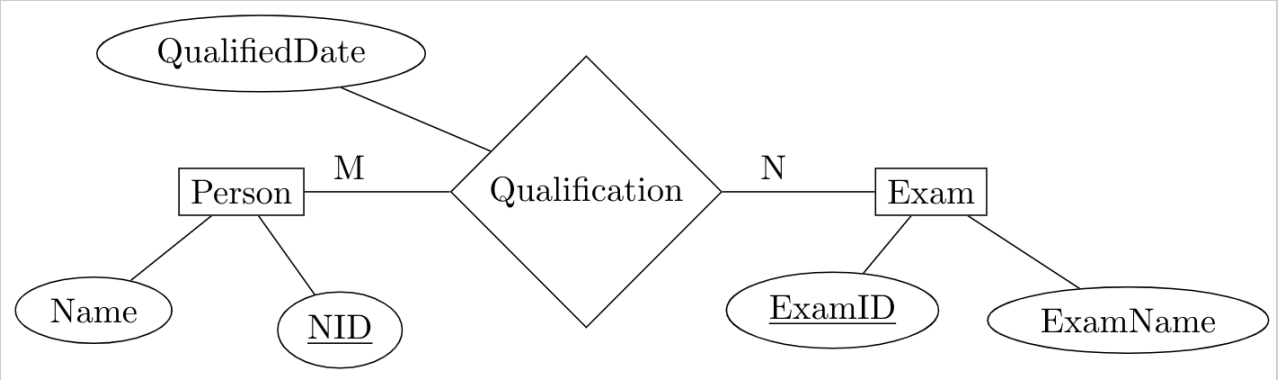
We want to convert the following diagram to the relational model. Choose the MOST appropriate answer from the following.



Select one alternative:

- PART(Part_no, Description), MANUFACTURED_PART(Part_no, Manufacture_date, Batch_no, Drawing_no), PURCHASED_PART(Part_no, Supplier_name, List_price) ✔
- PART(Part_no, Description, Manufacture_date, Batch_no, Drawing_no, Supplier_name, List_price)
- PART(Part_no, Description), MANUFACTURED_PART(Manufacture_date, Batch_no, Drawing_no), PURCHASED_PART(Supplier_name, List_price) ✖
- MANUFACTURED_PART(Part_no, Manufacture_date, Batch_no, Drawing_no, Description), PURCHASED_PART(Part_no, Supplier_name, List_price, Description)

7 ER to relational model



Which of the following relations will NOT hold if the above ER diagram is mapped into a relational model?

Select one alternative:

- Exam (ExamID, ExamName)
- Person (NID, Name)
- Exam(ExamID, NID, ExamName)
- Qaulification (NID, ExamID, QualifiedDate)

Consider the following populated database, with the two tables CAR and OWNER where VIN is the primary key for CAR and OwnerID is the primary key for OWNER. Also, CAR.OwnerID is a foreign key to OWNER.OwnerID.

CAR				
VIN	Year	Manufacturer	Color	OwnerID
1234	2001	Toyota	Red	1
4201	2017	Tesla	Silver	2
5678	2005	Ford	Black	3
9999	1999	Saab	Blue	1
1111	2010	Volvo	Green	NULL
2345	2015	Volvo	Gray	3

OWNER		
OwnerID	Name	Age
1	Mandy	19
2	George	45
3	Styrbjörn	39
4	Ingrid	80

13 **SQL**

Which of the following SQL statements will return the number of cars owned by the owner with OwnerID 4?

- SELECT COUNT(VIN)
  FROM CAR.OwnerID = 4;

- SELECT COUNT(*)
  FROM CAR, OWNER
  WHERE OWNER.OwnerID = 4;

- SELECT COUNT(*)
  FROM OWNER
  WHERE OwnerID = 4;

- SELECT COUNT(VIN)
  FROM CAR
  WHERE OwnerID = 4;

9 SQL

Which of the following SQL statements will return all information available in the CAR table about both red and black cars?

- (selected) SELECT *
  FROM CAR
  WHERE Color = 'Red' OR Color = 'Black';

- SELECT
  FROM CAR
  WHERE 'Red' OR 'Black';

- SELECT *
  FROM CAR
  WHERE Color IS 'Red' OR 'Black';

- SELECT *
  FROM CAR
  WHERE Color = 'Red' AND Color = 'Black';

8 SQL

Which of the following SQL statements will return the number of cars that have an owner?

- SELECT COUNT(*)
  FROM OWNER;

- SELECT DISTINCT(COUNT NAME)
  FROM CAR, OWNER
  WHERE OWNER.OwnerID = CAR.OwnerID;

- SELECT COUNT(DISTINCT Name)
  FROM CAR, OWNER
  WHERE CAR.OwnerID = OWNER.OwnerID;

- SELECT COUNT(*)
  FROM CAR
  WHERE OwnerID IS NOT NULL;

19 SQL

Which of the following SQL statements will return the number of cars that are Volvos?

- SELECT COUNT(*)
  FROM CAR
  WHERE CAR.Manufacturer IN (CAR SELECT 'Volvo');

- SELECT COUNT('Volvo')
  FROM CAR.Manufacturer;

- SELECT Number
  FROM COUNT(CAR.Manufacturer = 'Volvo') AS Number;

- SELECT COUNT(*)
  FROM CAR
  WHERE CAR.Manufacturer = 'Volvo';

12 **SQL**

We want to get a list of the names of owners of cars manufactured after 2004. Which of the following SQL-statements is correct?

- (selected) SELECT DISTINCT Name
  FROM OWNER, CAR
  WHERE CAR.OwnerID = OWNER.OwnerID AND CAR.Year > 2004;

- SELECT DISTINCT Name
  FROM CAR
  WHERE Year > 2004;

- SELECT DISTINCT Name
  FROM OWNER
  WHERE OWNER.OwnerID = CAR.OwnerID AND CAR.Year > 2004;

- SELECT DISTINCT Name
  FROM CAR LEFT OUTER JOIN OWNER ON CAR.OwnerID = OWNER.OwnerID
  WHERE Year > 2004;

Consider the following database

T1

C1	C2	C3
a	x	1
a	y	5
b	z	2
c	z	2
d	u	3

T2

C1	C2	C3
a	z	4
a	x	3
b	z	2
b	x	1
c	u	2

24 SQL B1

What is the result of the following SQL query?

```
SELECT count (T1.C1)
FROM T1
WHERE T1.C2 IN
      (SELECT T2.C2 FROM T2
       WHERE T2.C2="u");
```

Select one alternative

- The statement is incorrect
- An empty table
- 2
- None of them is correct.
- 7
- 1

20 **SQL B1**

Consider the following SQL statement and SQL results:

SELECT DISTINCT T1.C1
FROM T1, T2
WHERE T1.C2=T2.C2;

A	B	C	D
a	a	x	a
a	b	y	b
b	c	z	c
b		u	d
c			

Select one alternative:

- Result A is correct
- Result B is correct
- Result D is correct
- Result C is correct

21 **SQL B1**

Consider the following SQL statement and SQL results:

```
SELECT T1.C2, SUM(T1.C3*T2.C3)
FROM T1, T2
WHERE T1.C2 = T2.C2
GROUP BY T1.C2
HAVING SUM(T1.C3*T2.C3)>5;
```

A

z	24

B

y	35

C

x	7
y	35
z	10

D

u	6
z	24

Select one alternative:

- Result C is correct
- Result D is correct
- Result A is correct
- Result B is correct

22 **SQL B1**

Consider the following SQL statement and SQL result.

SELECT T1.C1
FROM T1 **INNER JOIN** T2 ON (T1.C1 = T2.C1)
WHERE (T1.C3>2);

A	B	C	D
a	a	x	a
a	a	y	a
b			NULL
b			NULL

Select one alternative:

- Result B is correct
- Result C is correct
- Result D is correct
- Result A is correct

26

♥ Functional Dependencies

The following table shows the current state of a relation, where C_1 is the **primary key** and C_2 is a **candidate key** of the relation. Which of the following options are correct? (Select **all** correct choices)

C_1	C_2	C_3	C_4	C_5
1	101	George	Anna	100
2	301	George	Chrisanthos	100
3	311	George	Iliada	100
4	401	John	Nektarios	2
5	891	John	Maria	1
6	999	Marina	Anna	3

♥ Select one or more alternatives:

- [ ] We cannot be certain whether the functional dependency $C_2 \rightarrow C_3$ holds.
- [x] The functional dependency $C_1 \rightarrow C_3$ holds. ✔
- [x] The functional dependency $\{C_3, C_4\} \rightarrow C_5$ could hold but we can not be sure for it. ✔
- [ ] The functional dependency $C_3 \rightarrow C_5$ could hold.
- [x] The functional dependency $C_2 \rightarrow C_4$ certainly holds. ✔
- [x] The functional dependency $C_2 \rightarrow C_1$ certainly holds. ✔

28 Normal Forms Q2 (DBI 2022)

Consider a relation R1(A, B, C, D), where {A, B} constitute the primary key and the following full functional dependencies hold:

- B → D
- C → D

and R2(G, E, F) where G is the primary key.

Select one alternative:

- R1 is in second normal form
- R1 is in third normal form
- We are sure that R2 is in second normal form. ✔
- None of the other is correct

27 Normal Forms Q1

Consider a relation R(A, B, C, D), where {A, B} constitute the primary key and the following full functional dependency holds:

- C → D

Which of the following holds?

- R is in the second normal form. ✔
- R is certainly in the third normal form but not in the first normal form.
- R is certainly in the second normal form but not in the first normal form.
- R is in the third normal form.

30 ♥ Serializability

For the following sets of transactions T_1, T_2, and T_3, which of the schedules are (conflict) serializable?:

♥ Select one or more alternatives:

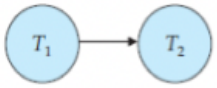
T1	T2	T3
	read_item(A);	
	A := A + 10	
	write_item(A);	
read_item(A);		
A := A * 3;		
write_item(A);		
read_item(B);		
B := B – 5;		
write_item(B);		
		read_item(C);
		C := C / 5;
		write_item(C)
		read_item(A)
		A := A / 3;
		write_item(A)

T1	T2	T3
read_item(A);		
		read_item(C);
	read_item(A);	
A := A * 5;		
write_item(A);		
read_item(B);		
B := B – 10;		
write_item(B);		
	A := A + 10	
		C := C / 10;
		write_item(C)
	write_item(A);	
		read_item(A)
		A := A / 5;
		write_item(A)

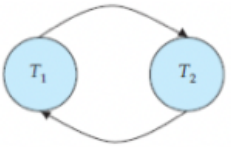
T1	T2	T3
read_item(A);		
A := A * 5;		
write_item(A);		
	read_item(A);	
read_item(B);		
B := B – 10;		
write_item(B);		
	A := A + 10	
		read_item(C);
		C := C / 10;
		write_item(C)
	write_item(A);	
		read_item(A)
		A := A / 5;
		write_item(A)

T1	T2	T3
	read_item(A);	
	A := A + 10	
		read_item(C);
		C := C / 5;
		write_item(C)
		read_item(A)
read_item(A);		
A := A * 3;		
write_item(A);		
read_item(B);		
B := B – 5;		
write_item(B);		
	write_item(A);	
		A := A / 3;
		write_item(A)

32 Transactions- Precedence graphs



Precedence Graph A1



Precedence Graph B1

Consider these two Precedence graphs, where T1 and T2 represent transactions. Which of the following is true?

Select one alternative:

- Precedence graph A1 corresponds to a Schedule that may suffer from the lost update problem.
- None of the other answers is correct!
- Precedence graph B1 corresponds to a Schedule that may suffer from the lost update problem.
- Both Precedence graphs, A1 and B1, corespond to schedules that are serializable.

33 **Recoverability**

Our banking application crashed. At the time of the crash (just after operation 9), the system log had the following entries:

1. begin_transaction(t1)
2. read_item(t1, Mike.Balance)
3. write_item(t1, Mike.Balance, 600, 550)
4. begin_transaction(t2)
5. read_item(t2, Jane.Balance)
6. write_item(t2, Jane.Balance, 400, 200)
7. read_item(t1, Alice.Balance)
8. write_item(t1, Alice.Balance, 800, 850)
9. commit(t2)

Based on the log, which of the following actions might a DBMS's recovery subsystem perform?

Select one alternative:

- Undo operations 3 and 8, possibly redo operation 6 ✔
- Undo all write operations (operations 3, 6 and 8)
- Undo operation 6, possibly redo operations 3 and 8
- Redo all write operations (operations 3, 6 and 8)
- Leave the database as it is

Consider the database below. Where, the primary keys are underlined and the foreign key (FK) relationships are:

- PROJECT.Dnum is FK ref. DEPARTMENT.Dnumber;
- PROJECT.Plocation is FK ref. DEPT_LOCATIONS.Dlocation;
- DEPT_LOCATIONS.Dnumber is FK ref. DEPARTMENT.Dnumber;
- DEPARTMENT.Mgr_ssn is FK ref. EMPLOYEE.Ssn;
- EMPLOYEE.Super_ssn is FK ref. EMPLOYEE.Ssn;
- EMPLOYEE.Dno is FK ref. DEPARTMENT.Dnumber.

Suppose each of the following update operations is applied directly to this database:

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

34

♥ Constraint violations

Choose the correct statements regarding the integrity constraints (i.e. key constraints, entity and referential integrity constraint) of the following operation when applied on the database shown to the left.

DELETE FROM EMPLOYEE WHERE Lname = 'Borg'

♥ Select one or more alternatives:

- [ ] It violates the key constraint.
- [ ] No constraint violations.
- [ ] It violates entity integrity constraint.
- [x] It violates referential integrity constraint.

35

♥ Constraint violations

Choose the correct statements regarding the integrity constraints (i.e. key constraints, entity and referential integrity constraint) of the following operation when applied on the database shown to the left.

DELETE FROM EMPLOYEE WHERE EMPLOYEE.SSN = '888665555'

♥ Select one or more alternatives:

- [x] It violates referential integrity because of the foreign key from DEPARTMENT to EMPLOYEE
- [x] It violates referential integrity because of the foreign key from EMPLOYEE to EMPLOYEE
- [ ] It violates the key constraint because the primary key of the deleted employee is equal to Mgr_ssn for the department with Dnumber 1.
- [ ] It violates entity integrity constraint because Super_ssn is NULL for the employee with Ssn 888665555.
- [ ] No constraint violations.
- [ ] It violates the entity integrity constraint because we are deleting based on a primary key.

39 Multilevel Security

A database with high security requirement employs multilevel security. In such a system, each element in a tuple has associated security level (**T**op **S**ecret (TS), **S**ecret (S), **C**onfidential (C), **U**nclassified (U)) corresponding to the sensitivity of the data stored. In some models, **T**uple **C**lassification attribute TC is added to the relation attributes. The value of TC in each tuple *t*—which is the highest of all attribute classification values within *t*—provides a general classification for the tuple itself.

Suppose the following state of the EMPLOYEE table.

EMPLOYEE table

Name	Salary	JobPerformance	TC
Paul **U**	20000 **C**	Average **C**	**C**
Steve **C**	40000 **C**	Good **S**	**S**
Jane **S**	80000 **S**	Excellent **S**	**S**

Choose the table which can be returned as response from the options below assuming the user has security level **U**:

Option 1

Name	Salary	JobPerformance	TC

Option 2

Name	Salary	JobPerformance	TC
Paul **U**	20000 **C**	Average **C**	**C**

Option 3

Name	Salary	JobPerformance	TC
Paul **U**	NULL **U**	NULL **U**	**U**

Option 4

Name	Salary	JobPerformance	TC
Paul **U**	20000 **C**	Average **C**	**C**
NULL **U**	NULL **U**	NULL **U**	**U**

Option 5

Name	Salary	JobPerformance	TC
Paul **U**	20000 **C**	Average **C**	**C**
Steve **C**	40000 **C**	Good **S**	**S**
Jane **S**	80000 **S**	Excellent **S**	**S**

Select one alternative:

- Option 3 ✓
- Option 5
- Option 4
- Option 1
- Option 2

40 SQL Injection

In a database application, the following fragment of code handles the search of a product.

```
cursor.execute(
   "SELECT * FROM PRODUCT
    WHERE Pname=' " + userInput + " ')
```

More precisely, the user enters the name of the product that they are searching for which is stored in the **userInput** variable. Which of the following user inputs can cause the most harm?

Select the input which can cause the most harm out of the following:

- ; DROP PRODUCT;
- DROP PRODUCT; --
- bag'; DROP PRODUCT;
- bag'; DROP PRODUCT; -- ✓

41 Choosing the right index

Given a table EMPLOYEE(ID, Fname, Lname, Salary, MgrID), where ID is the primary key and MgrID is a foreign key. We need to improve the performance of the following query:

```
SELECT Lname, Salary
FROM EMPLOYEE
WHERE Fname = "George"  AND Salary > 40000  AND MgrID = 1
```

For this purpose we want to use only one index. Which of the following indexes will not be useful at all?

Select one alternative:

- Hash index on Salary ✔
- Tree index on Fname
- Tree index on MgrID
- Hash index on MgrID
- Hash index on Fname